PRD: Improving ManageXR's Analytics Feature

(Lan Nguyen – April 2025)

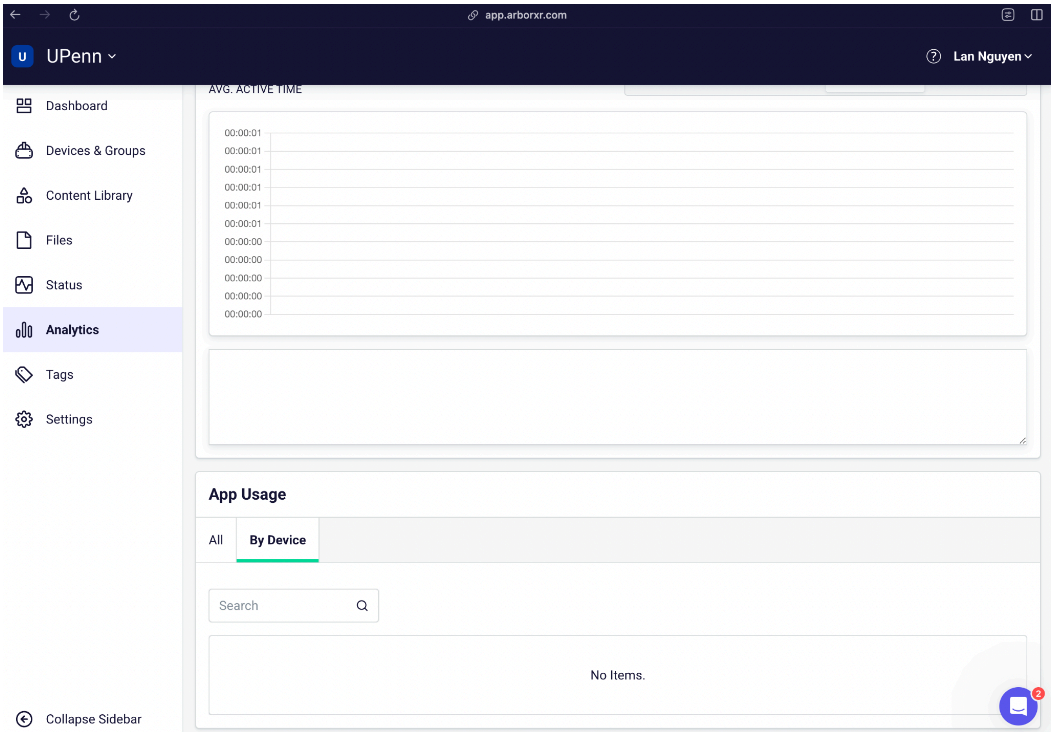
Prioritization

After reviewing the three opportunities (improving the Analytics feature, building a ManageXR mobile app, and building the Kiosk Videos Playlist), I would choose to prioritize the Analytics feature first for the following reasons:

1. **Direct alignment with our North Star metrics:** Devices on Platform and Monthly Active Devices. By making it easier to track single devices and get answers to specific questions, customers will be more confident adding devices to the platform and keeping them active. If administrators can't find insights for their current devices, they're less likely to add more devices to their fleet.
2. **Implementation efficiency:** Compared to developing a new mobile app, enhancing our existing analytics requires significantly less development effort as the infrastructure for charts already exists, minimizing engineering lift compared to new mobile development.
3. **High reach and visibility feature**: Analytics is an extremely foundational experience to the platform, whose improvements will be seen by and affect all customers.
4. **Strategic alignment:** At this stage in the market, ManageXR should focus on improving the core product experience rather than expanding to new platforms like

mobile, which, despite API reuse, would still require significant effort to develop mobile-specific UI components.

5. **Competitive necessity:** A quick competitive analysis indicates that direct competitors like ArborXR are already offering device-level analytics, making this a table-stakes feature we need to implement to remain competitive in the market.



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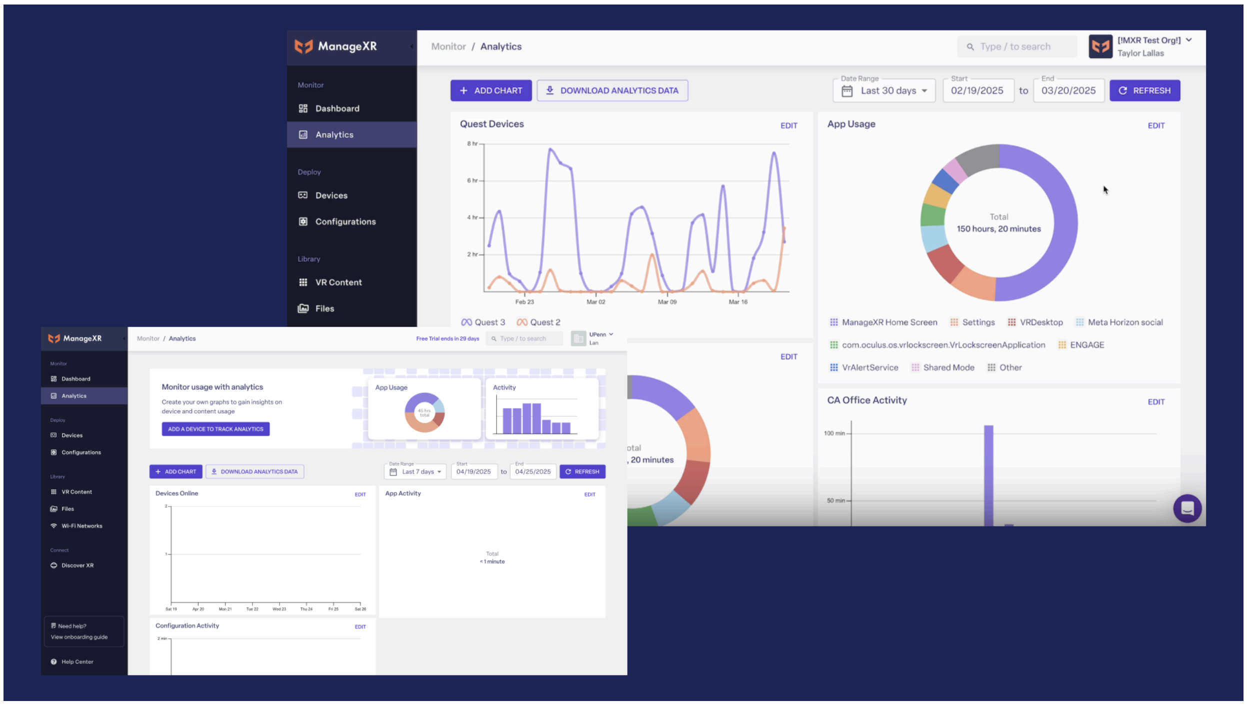
Proposed Improvements

Background & Motivation



Analytics is a foundational feature of the ManageXR platform (second to Dashboard on the nav bar), serving as the primary window through which customers monitor, understand, and optimize their XR deployments. As organizations scale their XR initiatives from pilot programs to enterprise-wide deployments, robust analytics becomes increasingly critical to their success and, by extension, to ManageXR's growth toward our North Star metrics of total devices and monthly active devices.
The current Analytics feature, as observed in the platform today, offers a set of fundamental visualization tools:

- A dashboard-style interface with multiple chart widgets, most notably in terms of device usage time, total app usage, total configurations activity
- Chart edit and customization options (chart type, unit of measure)
- Date range selection for filtering data (ie, last 30 days, custom range)
- Basic download functionality for analytics data in CSV

While the above capabilities provide value for high-level oversight, the implementation has significant limitations. The current design focuses primarily on aggregate data across entire device fleets, with no way to drill down to individual device performance. The charts are relatively static, offering limited ability to answer specific business questions that drive operational decisions.

Problem statement

Customers cannot effectively monitor individual devices or extract actionable insights from their XR fleet data.

- Current analytics only show total data across device fleets, making it difficult to understand the performance of specific devices
- Customers struggle to answer key business questions (like "Which devices haven't been used this month?" or "What apps are most/least popular?")

Target use cases

1. As an XR tech administrator, I want to see detailed metrics for a specific device so I can effectively troubleshoot issues.
2. As a fleet manager, I want to quickly identify inactive devices so I can redistribute or troubleshoot them.
3. As a training coordinator with limited time and technical resources, I want to know quickly which apps are most and least used so I can focus resources effectively.

Current user journeys

1. **Troubleshooting a problematic device:** User navigates to Analytics > Sees only aggregate data > Cannot find individual device metrics > Contacts customer support or manually checks device
2. **Identifying inactive devices:** User navigates to Analytics > Views general usage charts > Cannot determine which specific devices are inactive > Manually checks device, or manually creates a spreadsheet tracking system
3. **Determining application popularity:** User navigates to Analytics > Sees general app usage data > Cannot determine which apps are most/least used across specific devices or configurations > Makes decisions based on incomplete information

Objectives

Goals

- Enable device-level analytics to help customers identify and resolve single-device issues more quickly
- Help users answer specific questions of interest: Enable quick and easy data-driven decision making by creating pre-built reports that address common insights/questions.
- Maintain the simplicity and ease of use of the current UI while adding more powerful capabilities: Ensure the enhanced analytics remain accessible to non-technical users.

Non-goals/Out-of-scope

- Data export and API integration capabilities (should be a separate future proposal)
- Custom report builder functionality
- Mobile app analytics features
- Changes to the overall analytics navigation structure

Requirements

P0 = Must-have, P1 = Should-have, and P2 = Nice-to-have

Priority	Detailed description
P0	**Users can switch between Total Aggregate data Overview and Single Device data** • Create a prominent toggle between "Overview" and "By Device" tabs on the main Analytics dashboard • Overview tab displays data aggregated across all devices (existing functionality) • By Device tab displays a list of all connected devices • Ensure clicking Analytics in the main navigation bar leads to the Overview dashboard by default
P0	**Users can view key performance metrics at a glance** • Add quick insights panels to both the Overview and Device Detail dashboards • Generate insights about user-customized metrics with period-over-period comparison (such as inactive devices, popular apps, usage trends) • Make insights clickable to deep-link to relevant detailed reports • Update insights daily based on latest data

Priority	Detailed description
P0	**Users can view a filterable list of all devices, each comes with its own data and insights** • Create a searchable table with all devices in the fleet • Reuse existing table components from the "Devices" feature for consistency • Include search field • Add column filtering and sorting • Allow bulk selection of devices for batch operations • Table displays “Status,” “Last Active,” “Usage,” “Configuration Activity”
P0	**Users can access comprehensive analytics for a specific device** • Create a dedicated device analytics page accessible by clicking device name in the list view • Include a back button to return to the device list view • Display device name prominently as the page title
P1	**Users can access common device management tasks from the analytics interface** • Add action buttons for quick management of inactive devices • Provide direct links to device configuration pages where appropriate
P1	**Users can create device-specific charts** • Maintain consistent chart creation experience from Overview analytics • Allow adding custom charts specific to this device • Enable editing and customization of charts
P2	**Users can customize the analytics dashboard** • Allow reordering of insight panels based on importance to the user • Enable saving of custom dashboard layouts
P2	**Users can export the device list** • Enable exporting the filtered device list to CSV/Excel • Allow scheduled exports of device status reports
P2	**Users can compare device performance against fleet averages** • Add benchmark indicators comparing to fleet averages • Highlight significant deviations from expected usage patterns

Technical Implementation Considerations

The implementation will leverage our existing React/TypeScript frontend and NestJS backend. New API endpoints will be needed for:

- Device-specific metrics retrieval
- Aggregated report data
- Insight generation algorithms

We expect to reuse much of our existing charting and UI components to maintain visual consistency.

Success Metrics

Quantitative

- Increased monthly user engagement with Analytics
- Increased monthly active devices
- Increased total devices on platform

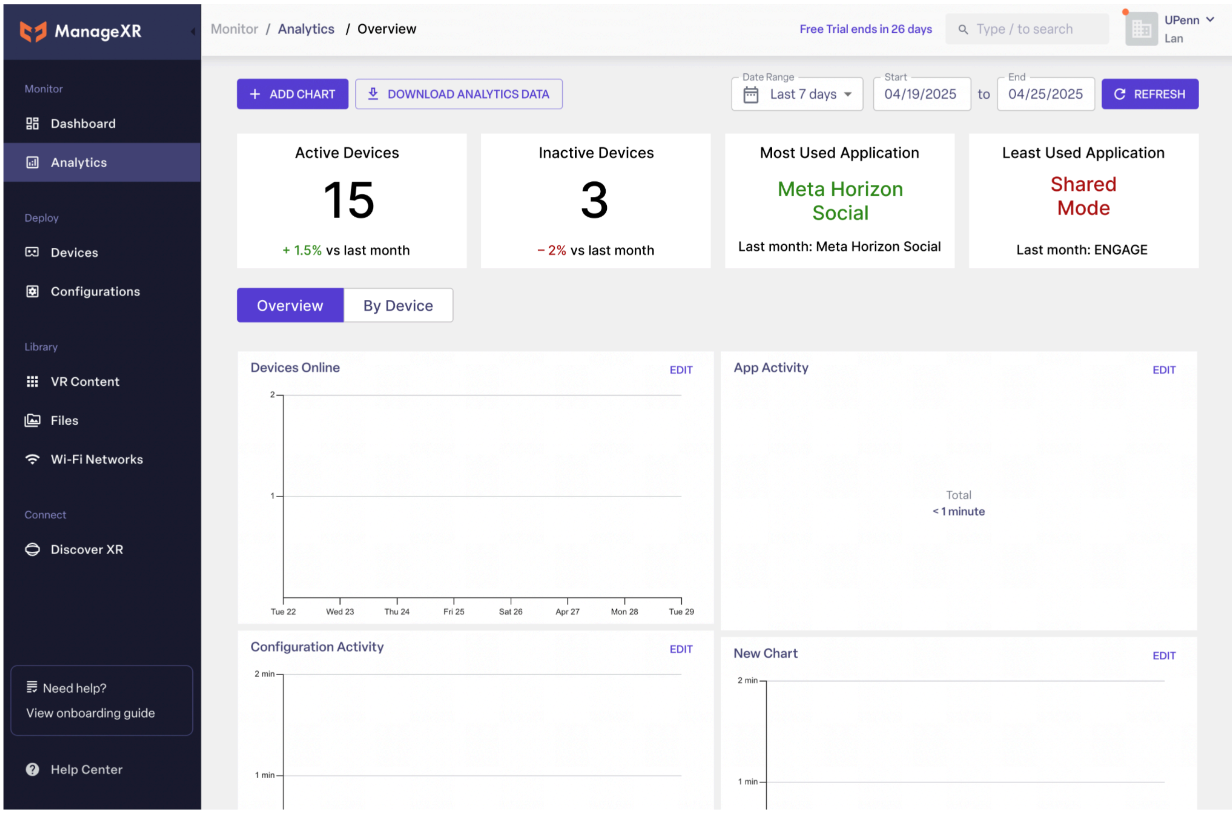
Qualitative

- Improved customer satisfaction (user survey, NPS feedback)
- Increased time savings for admins in pulling data insights (measured by user surveys)

Wireframes

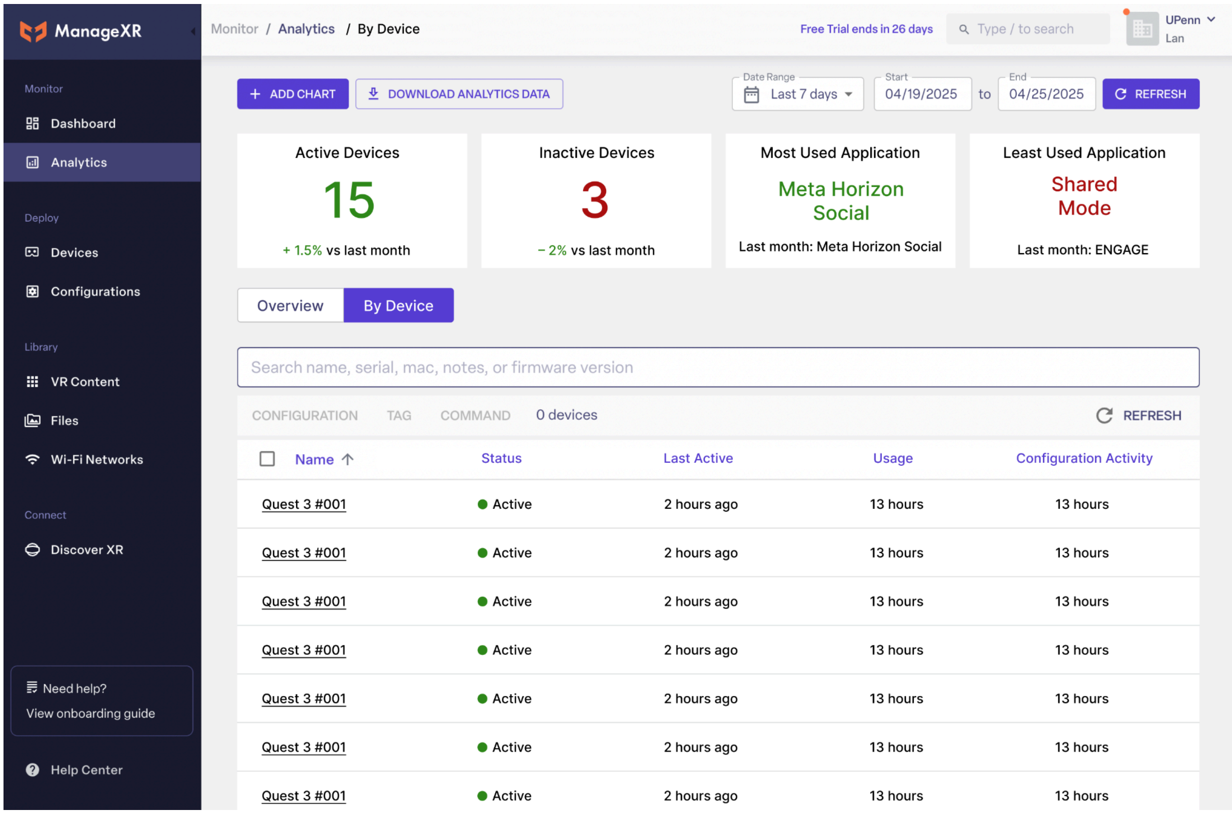
1. New Analytics Dashboard Page

- Quick Insights panels prominently displayed
- Clear navigation to By Device list view
- Maintains existing overview total analytics charts for continuity
- Users can edit what information they would like to be displayed in Quick Insights panels



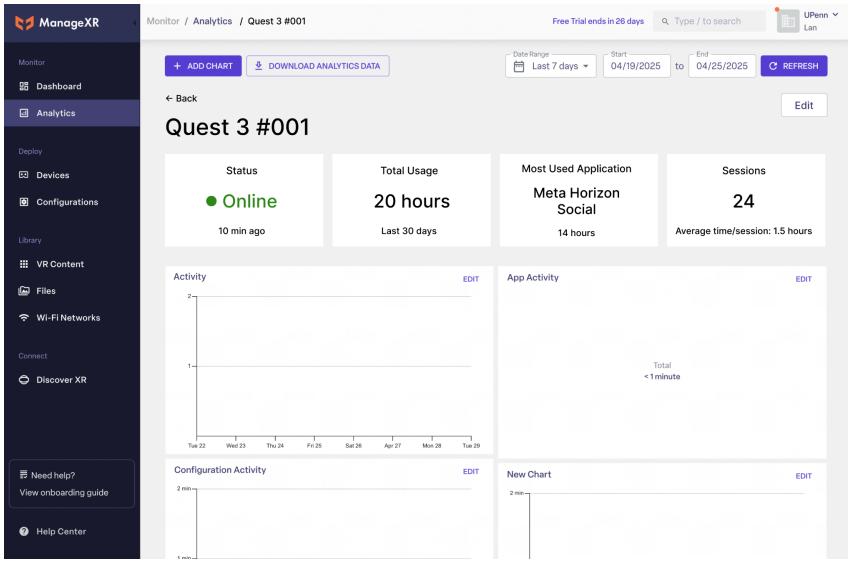
2. Analytics By Device List View

- Reused table components from “Devices” feature
    - Searchable and filterable table of devices
- Status indicators with clear visual design
- Sortable columns for different metrics
- Links to device detailed analytics pages (by clicking device name)



3. Single-Device Analytics Details

- Quick device insights summary panels
- Ability to create charts for specific insights/device just like Overview
- Back button goes to Device List view



Risks and mitigations

Risk	Impact	Likelihood	Mitigation
Low customer adoption	High	Low	Create in-app tours highlighting new features, add tooltips explaining value, and prepare customer success team with talking points.
Data accuracy issues for offline devices	Medium	High	Clearly indicate data freshness timestamps, implement sync status indicators, and document data collection methodology.
Development complexity exceeds estimates	Medium	Medium	Phase development to deliver core single device view first, then add Quick Insights incrementally. Create detailed technical specs before implementation begins.

Assumptions

There were several assumptions I made in the making of these requirements since I did not have access to the company's detailed business data, user research, or an actual XR device to test out all the available Analytics features (among others):

- I had to guess the insights that I thought users would be most interested in for the Quick Insights panels, along with each respective data and numbers that would seem the most reasonable
- I assumed we can retrieve all necessary device metrics through the existing device APIs
- I assumed most customers will be primarily interested in 30-day timeframes for metrics
- I assumed we do not need to implement real-time data updates (current refresh rate is sufficient